MILITARY SPECIFICATION

DECONTAMINATING AGENT, BIOLOGICAL, ETHYLENE OXIDE-PROPELLANT MIXTURE

1. SCOPE

1.1 This specification covers one type of biological decontaminating agent consisting of an ethylene oxide-propellant mixture.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

BB-C-310 - Chlorofluoro Hydrocarbons (of the Methane and Ethane Series)

FSC 6850

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MIL-E-52171 - Ethylene Oxide, Technical

STANDARDS

MILITARY

MIL-STD-129 - Marking for Shipment and Storage

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING MATERIALS

ASTM Standards

D1364 - Method of Testing for Water in Lacquer Solvents and Diluents

(Applications for copies of the ASTM publications should be addressed to the American Society for Testing Materials, 1916 Race St., Philadelphia, Pennsylvania.)

CODE OF FEDERAL REGULATIONS

49 CFR 71-90 - Interstate Commerce Commission Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles.

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(The Interstate Commerce Commission Regulations are a part of the Code of Federal Regulations available from the Superintendent of Documents, Washington 25, D. C. Orders for the above publication should cite "49 CFR 71-90".)

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules

(Application for copies of these freight classification rules should be addressed to the Uniform Classification Committee, 202 Union Station, Chicago 6, Ill.)

3. REQUIREMENTS

3.1 Chemical composition. The chemical composition of the decontaminating agent shall consist of an ethylene oxide-propellant mixture as follows:

3.1.1 Ethylene oxide. Technical ethylene oxide conforming to Specification MIL-E-52171 shall constitute 11.0 ± 0.5 percent by weight of the mixture when tested as specified in 4.5.1.

3.1.2 Propellant. The propellant making up the remainder of the misture shall consist of equal parts by weight, within a tolerance of ± 1.0 percent, of trichloromonofluoromethane (type 11) and dichlorodifluoromethane (type 12) conforming to Specification BB-C-310.

3.2 Water content. The water content of the agent shall not be more than 0.05 percent when tested as specified in 4.5.2.

3.3 Contamination. The agent shall be water white and free from polymers, sediment, dirt, metal particles, or other foreign matter when tested as specified in 4.5.3.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Special provisions.

4.2.1 Alternative inspection (including testing) procedures. The supplier may utilize any alternative inspection procedure which will insure equal or better quality by submitting a written proposal with justification and obtaining written approval from the Government prior to instituting the procedure. In case of dispute, the procedures of this specification will govern.

4.2.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1.2 and section 5 have been satisfied.

4.3 Lotting. A lot shall consist of the decontaminating agent produced by one manufacturer from the same materials and under essentially the same manufacturing conditions.

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4.4 Sampling. Sample containers shall be taken at random from each lot as follows:

Lot size	Sample size
1	1
2- 275	2
276- 545	3
546- 900	4
901-1345	5
1346-1875	6
1876-2500	7

Sample size for lots exceeding 2500 containers shall be calculated using the following equation:

$$n = 0.15\sqrt{N}$$

where: n = sample size.

N = lot size.

Each sample container shall be inverted or positioned on its side on an inclined support so that the valve end is low. Liquid samples shall be taken as follows and labeled to identify each sample with the lot and the container from which it was taken:

4.4.1 For the ethylene oxide content test. From each sample container an approximate 10-milliliter (ml) sample shall be drawn into a sampling device (see 6.2 and fig. 1) which has been evacuated to an absolute pressure of not more than 0.1 millimeters (mm) of mercury. The sample

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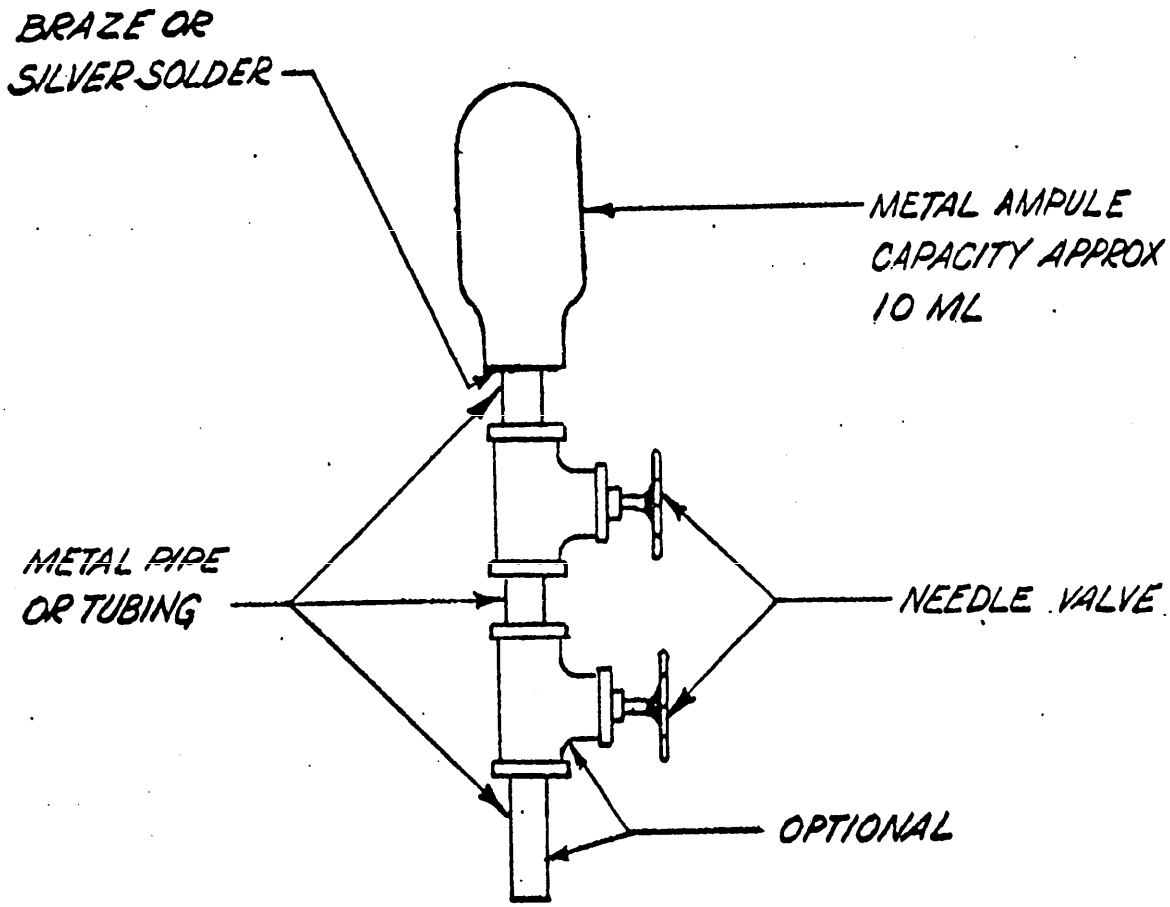


FIGURE -1 SAMPLING DEVICE

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shall be tested as specified in 4.5.1.

4.4.2 For the water content and contamination tests. A 100-ml sample shall be drawn from each sample container and placed in a chilled, clean, dry container. Samples shall be maintained at a temperature not higher than minus 30°C by means of a dry ice-acetone bath until the tests have been completed.

4.5 Tests. Analytical reagent grade chemicals and distilled water shall be used throughout the tests. Care must be exercised in preparing, handling, and testing the agent because ethylene oxide is a toxic material. (See Specification MIL-E-52171.) All tests shall be performed under a ventilated hood. Where applicable, blank determinations shall be conducted in parallel with the tests, and corrections applied when significant. Tests shall be conducted as follows:

4.5.1 Ethylene oxide content.

4.5.1.1 Reagents.

4.5.1.1.1 Magnesium bromide solution. Prepare a solution of 1.0 N sulfuric acid in a saturated solution of magnesium bromide ($MgBr_2$ or $MgBr_2.6H_2O$). (See 6.3).

4.5.1.1.2 Bromcresol purple indicator solution. Use a 0.04 percent bromcresol purple solution.

4.5.1.2 Procedure. Weigh the filled sampling device (see 4.4.1) to the nearest 0.02 gram (g), and record the weight as W_1. Secure the device in an upright position and attach the plastic tube of a delivery fixture (see 6.3 and fig. 2). Measure 50 ml of the saturated magnesium bromide

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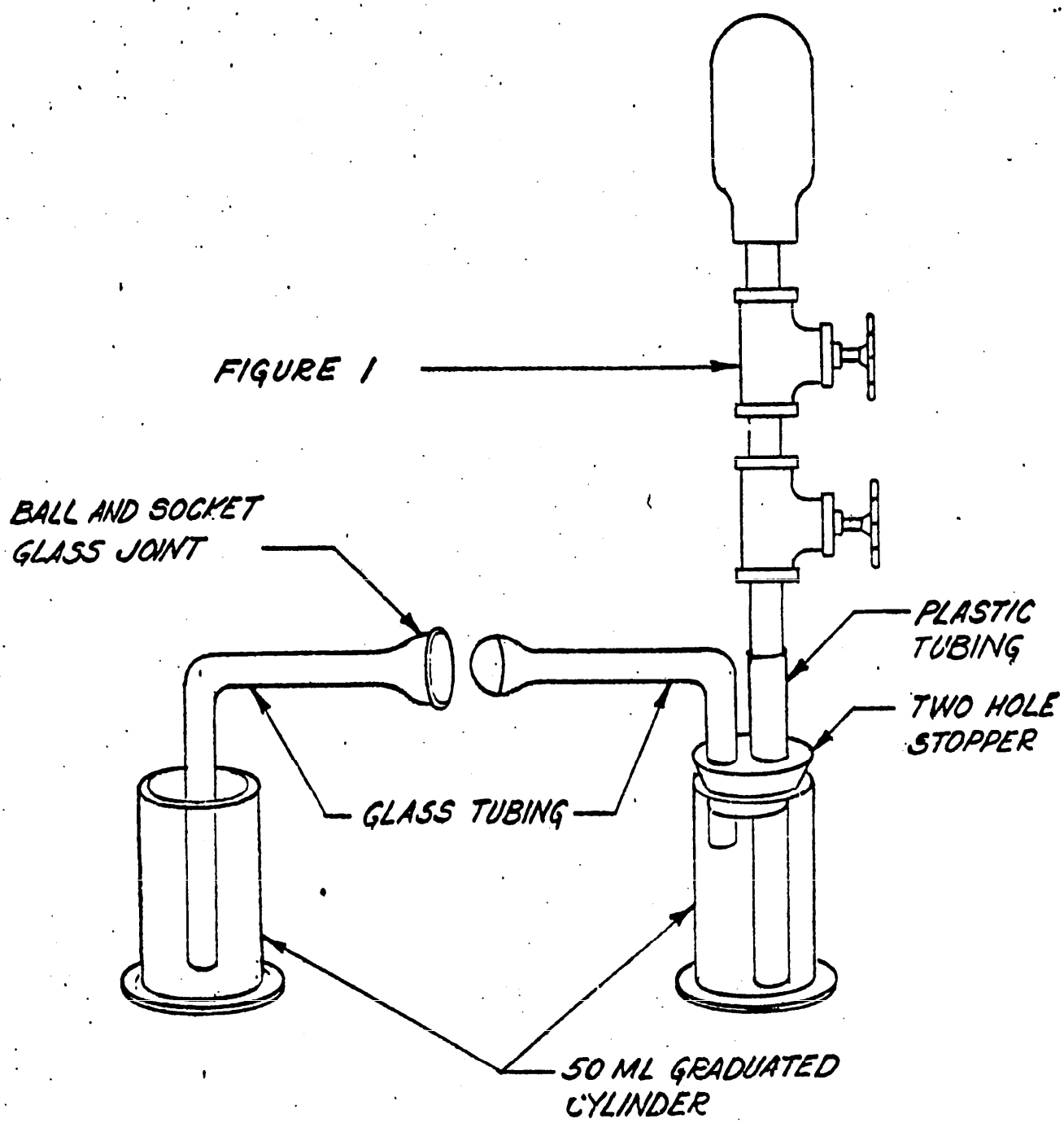


FIGURE 2 — DELIVERY FIXTURE

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reagent solution in a burette, drain 30 ml into one of the cylinders and 20 ml into the other cylinder. Stopper the cylinder containing 30 ml of solution. Insert the plastic tube through the stopper and into the cylinder so that the end of the tube is approximately 1/8 inch above the bottom of the cylinder. Position the glass tubing as shown on figure 2 and assemble the glass joint so as to provide a gas-tight connection. Open the upper needle valve slightly (if two valves are used) and control the flow of the agent by means of the lower valve so that five or six bubbles are visible in the first cylinder. Continue discharging the agent into the solution until not less than 8.0 g has been removed from the sampling device (30 to 45 minutes). Tightly close the lower needle valve and disassemble the delivery fixture. Remove the plastic tube from the sampling device and remove both tubes from the solution. Wash the tubes with a little water into one of the cylinders. Weigh the sampling device to the nearest 0.02 g and record the weight as W_2. Allow 10 minutes to complete the reaction and pour the solution from the two cylinders into a 250-ml flask, rinse the cylinders with 50 ml of water and add the rinse to the solution in the flask. Add three to five drops of bromcresol purple indicator to the solution and titrate with 1.0 N NaOH solution to the purple end point. Record the ml of 1.0 N NaOH solution as B.

4.5.1.3 Blank determination. Measure 50.0 ml of the saturated magnesium bromide reagent solution in a burette, drain into a 250 ml flask, and add 50 ml of distilled water. Add three to five drops of bromcresol purple

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indicator to the solution and titrate with 1.0 N NaOH solution to the purple end point. Record the ml of 1.0 N NaOH solution used as A.

4.5.1.4 Calculation. Calculate the percentage by weight of ethylene oxide in the agent as follows:

$$\text{Percent ethylene oxide} = \frac{4.405\ N(A-B)}{W_1 - W_2}$$

where: W_1 = initial weight of the sampling device.

W_2 = final weight of the sampling device.

A = ml of 1.0 N NaOH used for titration of the blank.

B = ml of 1.0 N NaOH used for titration of the agent.

N = normality of 1.0 N NaOH.

4.5.2 Water content. Determine the water content of the sample in accordance with ASTM Standard D1364 using a 50-ml portion of the sample (see 4.4.2). Calculate the weight of the amount used, by determining the specific gravity at minus $30^{\circ} \pm 1^{\circ}C$.

4.5.3 Contamination. Transfer the remainder of the sample to a nessler tube chilled to approximately minus 30°C for a period of 10 minutes and observe visually for compliance with 3.3.

4.6 Acceptance/rejection criteria. If any sample of the agent fails to meet the requirements of this specification, the lot represented shall be rejected.

5. PREPARATION FOR DELIVERY

5.1 Packing

5.1.1 Level A. The agent shall be packed in cylinders conforming to Interstate Commerce Commission Specification 4B, size as specified (see 6.1

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Cylinders shall be dry, clean, and free from residue of cleaning compounds or other foreign matter at time of filling.

5.1.2 Level C. The agent shall be packed in a manner to insure carrier acceptance and safe delivery at destination. Containers shall be in accordance with Uniform Freight Classification Rules or regulations of other carriers applicable to the mode of transportation.

5.2 Marking. In addition to any special marking required by the contract or order, all shipping containers shall be marked in accordance with Standard MIL-STD-129.

6. NOTES.

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Unit quantity required.

(c) Level of packing required.

6.2 Sampling device. A satisfactory sampling device (see fig. 1) has been made by joining one or two small needle valves in series to an empty carbon dioxide cylinder of appropriate size. The use of two valves will facilitate repetitive testing. The device should be capable of withstanding an internal pressure of 100 psig and a vacuum of 0.1 mm of mercury without leakage.

6.3 Delivery fixture. Plastic and rubber components of the delivery fixture (see fig. 2) should be checked frequently for deterioration. The corrosive effects of ethylene oxide as well as the danger of glass breakage may be eliminated by the use of corrosion-resisting steel as an

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alternate material, and appropriate changes in fixture design.

6.4 Magnesium bromide solution. Preparation of the magnesium bromide solution may be facilitated by first preparing a saturated solution of $MgBr_2.6H_2O$ in distilled water, and then adding sufficient diluted (one part acid, two parts water) sulfuric acid (H_2SO_4) to make a 1.0 normal solution. If necessary the final solution may be filtered through glass wool.

6.5 Supersession data. This specification supersedes Chemical Corps Purchase Description 199-1-23 dated 14 November 1961.

NOTICE. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whasoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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